

Weather Forecasting Project

This project leverages data science and visualization techniques to forecast weather conditions and present them in an interactive dashboard format using Power BI.

Project Files

- `Weather Forecasting.ipynb`: Jupyter Notebook containing the data preprocessing, analysis, and machine learning model for weather prediction.
- `Weather Dashboard.pbix`: Power BI report file with visualizations for weather data and prediction outputs.

Objectives

- Build a model to predict future weather conditions based on historical weather data.
- Clean and preprocess raw weather data to ensure accuracy.
- Train and evaluate machine learning models for forecasting.
- Visualize trends, predictions, and insights in a user-friendly Power BI dashboard.

Technologies Used

- **Python** (NumPy, Pandas, Matplotlib, Scikit-learn, etc.)
- **Jupyter Notebook**
- **Power BI** for reporting and dashboarding



Features

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Model building and accuracy evaluation
- Interactive Power BI dashboard for:
 - Temperature and humidity trends
 - Weather condition breakdown
 - Predictive insights



Insights

Some key visual insights from the Power BI dashboard include:

- Seasonal temperature and rainfall patterns
- Most frequent weather conditions
- Forecast trends and confidence scores



How to Run

1. Notebook

1. Install required Python packages.
2. Run `WeatherForecastingProject.ipynb` cell by cell.
3. View predictions and accuracy metrics.

Daily Temperature Range (2012-2015)

